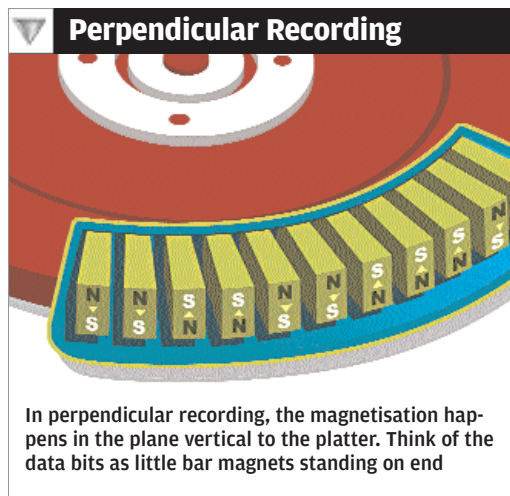




itself to denser packing together of magnetic clusters. According to a 2005 Hitachi GST (Global Storage Technologies) report, "the geometry and coercivity advantages of PMR led scientists to believe in potential areal densities that are up to 10 times greater than the maximum possible with longitudinal recording. Given current estimates, that would suggest an areal density using perpendicular recording as great as one terabit psi—making possible, in two to three years, a 3.5-inch disk drive capable of storing a terabyte of data."

Of course, it's not all rosy, and challenges remain. As an indicator, the same Hitachi GST report goes on to say, "Even though PMR is technically akin to the current generation of longitudinal devices, a number of technical challenges remain. For example, engineers are engaged in research to invent new kinds of read/write heads; to experiment with new materials that have enhanced magnetic properties and improved surface finishes; to maintain signal-to-noise ratios as the magnetic bits and signals become smaller; and to detect and interpret the magnetic signals using ever more advanced algorithms."

IDC, a global provider of market intelligence for the IT and telecom industries, said in a recent (2005) report that PMR technology will be key to the hard drive industry, and that its adoption cannot be delayed. "The transition to PMR and to the new heads and media required for this technology will begin late in 2005 and will be broadly adopted in products by the end of 2007." IDC predicts that PMR will be ubiquitous in the 630 million hard drives that will be shipped in 2009.

And yes, it's not all in the future. As far back as November 2002, Seagate Technology announced that its scientists had broken new ground by demonstrating areal densities of over 100 gigabits psi using PMR. That's already a tenth of what is predicted as being possible. And in July 2005, Seagate announced that virtually all its hard disk production would use PMR by the end of next year. "Over the coming year, the vast majority of products shipping (from Seagate) will be using perpendicular technology," said Charles Pope, Seagate's Chief Financial Officer. This is notable because no other company has made

**The transition to PMR and to the new heads and media required for this technology will begin late in 2005 and will be broadly adopted in products by the end of 2007**

such bold claims about such a rapid transition of production to PMR.

Seagate's first hard disk to use platters with perpendicular recording capability will be the 2.5-inch Momentus 5400.3 with capacity of 160 GB, and will ship in the winter of 2005.

### Heat-Assisted Magnetic Recording (HAMR)

Another recording technology that has been proposed and demonstrated as a way out of the superparamagnetism problem is HAMR. HAMR promises densities of 50 terabits psi—huge compared to the 1 terabit psi that's promised by PMR! The basic idea is quite simple: use a laser to heat the spot at which the recording is to be done—the heating makes the medium easier to magnetise.

To elaborate a little, heat-assisted recording involves producing a hot spot (usually with a laser) on the media, while data is written magnetically at the same time. The net effect is that when the media is heated, the coercivity, or field, required to write on the media is reduced. This makes it possible to write on high-coercivity media, which have higher stability against superparamagnetism. This writing can be done despite the magnetic write heads having limited fields.

Explained in a different way, HAMR entails using films that hold very tightly onto the magnetic orientation of each memory cell—so tightly that a laser must heat up each cell to be able to write data. Because the materials hold magnetic charges so well, memory cells can be packed together far more tightly than with less robust, conventional materials.

Seagate spokesperson John Paulsen said in 2002 that HAMR would be incorporated in products between 2007 and 2012. But guesses and predictions about this date vary, and it's not entirely clear when HAMR will be commercially mainstream.

Reality check: where does HAMR stand today? Seagate has already demonstrated the technology with prototype HAMR devices, but hasn't actually made a working, out-of-factory HAMR hard disk.

